FINAL REPORT

UNDERSTANDING DIVERSE SOCIO-ECOLOGICAL SYSTEMS

THEME: VULNERABILITY

COURSE CODE: ECCSS 05

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Acknowledgement

This report would not have been possible without the generous cooperation and honest responses of the fishing community of Vashi Gaon. We extend our sincere gratitude to all the community members who devoted their time to share their personal experiences and challenges in detail. Their insights have been invaluable in helping us to understand the real-world impacts of infrastructure development on traditional livelihoods.

We would like to extend special thanks to Mr. Harishchandra Sutar, the local fishing cooperative leader, for his unwavering support throughout the course of our fieldwork. His assistance in facilitating our entry into the community and providing crucial contextual information about the broader socio-economic impacts.

We also express our sincere gratitude to **Dr. Geetanjoy Sahu**, Dean of the School of Habitat Studies, whose support and encouragement laid the foundation for this field work. We extend our deepest thanks to the faculty members, **Dr. Praveen Kumar**, **Dr. Kamal Murari**, and **Dr. Unmesh Patnaik**, for their constant guidance, feedback, and encouragement throughout the field work. Their academic expertise and mentorship were instrumental in shaping our approach and perspectives.

We are equally thankful to all our classmates from the **ECCSS, Class of 2026**, whose collaboration, shared efforts, and peer support created a productive and engaging environment throughout the field work. The shared commitment to learning, teamwork, and perseverance made this experience even more rewarding and memorable.

This report is the result of many hands and minds working together. We are truly grateful to everyone who contributed in big and small ways.

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1. INTRODUCTION

The rapid development of Mumbai's infrastructure network is one of the most resolute urban development projects of India. The Mumbai Trans Harbor Link (MTHL) and Thane Creek Bridge-III are two projects that seek to enhance regional connectivity and ease congestion in one of the world's most populous city areas. Yet their ecological impact goes far beyond the physical presence of such massive developments and irreversibly transforms coastal and marine environments that have long supported traditional fishing societies.

The fishing community of Vashi Gaon, composed of Mumbai's original inhabitants (Koli and other traditional fishing communities), finds itself at the epicenter of this development-induced disruption. Their centuries-old relationship with Thane Creek's marine resources is being severed by construction activities that have transformed the creek's hydrological patterns, decimated fish breeding grounds, and created physical barriers to traditional fishing routes. The resulting decline in marine resources has triggered a cascade of socio-economic challenges, from widespread income loss to cultural disintegration and psychological distress.

Despite constitutional safeguards for customary resource users and global norms on development-induced displacement, compensation mechanisms have been found wanting in acknowledging the special character of losses incurred by non-landed communities whose livelihoods are based on common resources. This policy gap at the systemic level has resulted in the Vashi Gaon fishing community being engaged in a legal struggle for equitable recognition and compensation while trying to cope with fast-degrading environmental and economic conditions.

This report examines the vulnerability of the fishing community in Vashi Gaon resulting from these infrastructure projects, focusing on how their exposure to environmental changes, sensitivity of livelihoods, and adaptive capacity interact to create various dimensions of vulnerability.

1.1. Research Objectives

This study takes a deep dive into how urban infrastructure development impacts traditional resource-dependent communities in India, especially as cities rapidly change. The main goals of this research are to:

i. Identify and examine the ecological, economic, and social challenges that the fishing community faces due to infrastructure projects, using both numbers and personal stories to paint a full picture.

ii. Create a vulnerability index that measures the various aspects of risk within the community.

iii. Assess how well current compensation and support systems are working.

iv. Offer suggestions for fairer infrastructure development practices and targeted assistance for those affected.

This study contributes to the literature on the social impact of infrastructure development, with special reference to fishing communities. Based on a case study of Vashi Gaon, it provides empirical insights on how vulnerability is created among communities that are based on natural resources when subjected to human-induced changes in their environment. The study has policy implications for policymakers and development agencies when making more integrated infrastructure plans and more effective compensation mechanisms. Second, the approach to a vulnerability assessment that has been developed in this study can be employed as a model in making similar analysis in other contexts

1.2. Literature Review

The concept of vulnerability has been extensively discussed in both disaster risk reduction and climate change adaptation literature. According to Adger (2006), vulnerability is defined as "the state of susceptibility to harm from exposure to stresses associated with environmental and social change and from the absence of capacity to adapt." This definition recognizes three key components of vulnerability: exposure to stressors, sensitivity to their impacts, and adaptive capacity.

Infrastructure development projects and their impacts on local communities have been studied by numerous researchers. Cernea's (1997) Impoverishment Risks and Reconstruction model identifies eight key risks that affected communities face: landlessness, joblessness, homelessness, marginalization, food insecurity, increased morbidity, loss of access to common property resources, and community disarticulation. This framework is particularly relevant to understanding the multidimensional impacts on fishing communities.

In the specific context of fishing communities, Bavinck et al. (2014) note that displacement and livelihood disruption often lead to a phenomenon of "aquatic dispossession," where traditional users of aquatic resources lose access rights without adequate compensation or alternatives. This concept is especially pertinent to the case of Vashi fishers who have experienced restricted access to traditional fishing grounds.

A study conducted by Nayak et al. (2014) among fishing populations in India points out that infrastructure development tends to ignore traditional ecological knowledge and resource management systems, thereby increasing vulnerability further. In the same vein, Swapan and Gavin (2011) contend that in the Indian situation, compensation measures usually concentrate on land ownership, not really catering to the needs of individuals whose livelihood involves common property resources like waters for fishing.

Gender-differentiated impacts emerge as a consistent theme in the literature. Fernandes and Menon's (2023) mixed-methods study of four fishing communities in Maharashtra found that women fish vendors experienced disproportionate livelihood impacts, with 78% reporting complete loss of independent income following disruption to local fish catches. Unlike male fishers who could potentially seek employment in distant waters, women vendors remained tied to local markets due to domestic responsibilities and cultural constraints.

Picture 1. Briefing on the study objectives

2. STUDY AREA

2.1. Geographic Location

Vashi Gaon is strategically located on the eastern bank of Thane Creek in Navi Mumbai, Maharashtra. The settlement sits approximately 4 kilometers east of Mumbai's eastern shoreline, positioned at the coordinates 19.0682° N, 72.9914° E. The area forms part of the larger Mumbai Metropolitan Region (MMR) and falls under the administrative jurisdiction of the Navi Mumbai Municipal Corporation (NMMC).

Thane Creek, that vast estuarine complex, borders the west of the settlement and the principal fishing area of the inhabitants. Thane Creek is approximately 26 kilometers in length and 1.5 to 2 kilometers in breadth near Vashi Gaon. The estuarine habitat is characterized by:

- A complex network of tidal channels and mudflats exposed during low tide
- Extensive mangrove forests covering approximately 1,690 hectares along its banks
- A gradual salinity gradient that creates diverse microhabitats supporting varied marine species
- Seasonal variations in water flow influenced by monsoonal patterns, with freshwater influx peeking from June to September

The settlement is bordered by rapidly developing urban areas, including the commercial center of Vashi to the east and the emerging business district of Airoli to the north. This geographic positioning places Vashi Gaon at the intersection of traditional fishing waters and Mumbai's expanding infrastructure network, particularly the transportation corridors connecting Mumbai Island to the mainland.

2.2. Socio-Economic Profile

The fishing community of Vashi Gaon is a close-knit, multi-generational group whose socio-economic fabric is deeply intertwined with the Thane Creek ecosystem. The majority of residents belong to the Koli and Agri communities, recognized as the original inhabitants and traditional custodians of Mumbai’s coastal resources.

Vashi Gaon's population is predominantly engaged in fishing and related activities. The community majorly consists of traditional fishers who have practiced their trade for generations, therefore, developing specialized knowledge of local marine ecosystems, fish migration patterns, and fishing techniques. A recent TISS survey identified 1,637 fisherfolk residing in the 'zone of influence' of major infrastructure projects, including Thane Creek Bridge-III, with similar communities in neighboring villages also affected. Fishing remains the mainstay occupation, but the community structure includes boat owners, crew, net repairers, and a significant proportion of women engaged in fish vending and shore-based fishing. Women, who previously had earned a significant percentage of domestic income from shore-based fishing and fish vending, have been disproportionately affected. Being disconnected from the shore and the dwindling fish stock have prompted many to abandon fishing altogether and take up small-scale activities like selling home-cooked food or repairing and sewing garments. The shift has hastened economic precarity and eroded their economic autonomy.

The local fish market remains a central economic hub, but reduced supply and increased costs (e.g., fuel for longer detours) have squeezed profit margins. Social infrastructure, such as cooperatives like the Mariyayi Machhimar Sahkari Sanstha Maryadit, plays a key role in community organization and advocacy, particularly in ongoing legal efforts for fair compensation.

In summary, the socio-economic profile of Vashi Gaon's fishing community is characterized by declining incomes, occupational displacement, gendered impacts, and a growing disconnect between traditional livelihoods and the rapidly urbanizing landscape. These changes have heightened the community's vulnerability, making effective compensation and targeted support critical for their resilience and well-being.

2.3. Historical Context of Fishing in Vashi Gaon

Vashi Gaon's fishing community represents one of Mumbai's oldest continuous settlements, with archaeological and historical evidence suggesting fishing activities in the region dating back to at least the 14th century. The community comprises predominantly Koli fishers, recognized as the aboriginal inhabitants of Mumbai, whose cultural identity, religious practices, and social organization have evolved in symbiotic relationship with the creek ecosystem.

Traditionally, in the past, the Vashi Gaon fishing was marked by sophisticated ecological knowledge systems, such as:

- Cyclical seasonal harvesting based on fish migrations and spawning seasons
- Different fishing techniques intended to catch various species and ecological niches within the creek
- Community-based access right to certain fishing premises
- Intergenerational transmission of maritime skills, particularly useful in the creek's complex system of channels and mangrove forest
- Practices of traditional conservation which regulated harvesting during breeding periods

This old system of knowledge has provided communities with sustainable harvesting of resources for centuries well before industrialization and urbanization set in. The relationship this community has with their ecosystem transcends economic reason; it extends to deep-seated spiritual and cultural ties as well. Their annual festivals, rituals, and social gatherings revolving around the creek and its bounty show the strength of this affiliation.

Prior to the recent infrastructure developments, the fishing community of Vashi Gaon had established stable, if modest, livelihoods based on these traditional practices. The ecosystem of Thane Creek supported a diverse catch including prawns (kolambi), Bombay duck, pomfret, crabs, and various other fish species that were either sold in local markets or to wholesalers supplying to Mumbai's larger fish markets.

2.4. Overview of Bridge Construction Projects

2.4.1. Mumbai Trans Harbour Link (MTHL)

Commissioned in January 2023, the MTHL is India's longest sea bridge, spanning 21.8 kilometers across Thane Creek and connecting Mumbai with Navi Mumbai. Key features with direct impacts on fishing activities include:

- 16.5 kilometers of the bridge over water, supported by 1,500 foundation piles driven into the creek bed
- Construction that required dredging operations and the creation of temporary work platforms
- Restricted navigation zones around bridge piers that permanently block traditional fishing routes
- Increased maritime traffic that disrupts fish migration patterns and creates safety hazards for small fishing vessels

Environmental impact assessments for the MTHL acknowledged potential negative impacts on fishing communities but significantly underestimated their magnitude and failed to account for cumulative effects when combined with other development projects.

2.4.2. Thane Creek Bridge-III

The Thane Creek Bridge-III project, adjacent to two existing bridges, aims to ease traffic congestion between Mumbai and Navi Mumbai. Construction began in 2020 with significant implications for Vashi Gaon:

- Additional piling work that has further modified benthic habitats and creek hydrology
- Increased sedimentation that has accelerated the silting of traditional fishing channels
- Construction debris that has damaged fragile mangrove ecosystems serving as fish nurseries
- Narrowing of navigable channels, forcing fishers to take longer routes and increasing operational costs

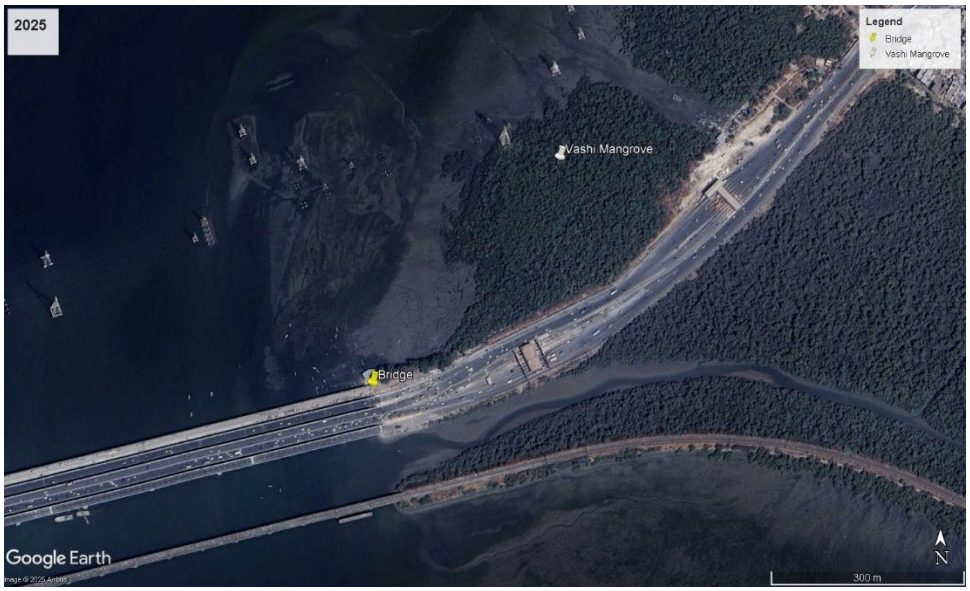
2.5. Cumulative Infrastructure Impact

In addition to these specific projects, the fishing village is also threatened by:

- Release of wastewater from neighboring industrial and residential developments
- Port development that has increased sea traffic and pollution
- Seaside road improvements that reorganized seaside access points
- Planned Special Economic Zones posing threats to existing fishing grounds

The combined effect of these trends is not cumulative, though, but synergistic, with each new project further lowering the ecological resilience of an increasingly stressed environment and leaving fishers with increasingly fewer and fewer options for adapting.

Picture 2. Vashi Creek in 2009



Picture 3. Vashi Creek today

Comparing the 2009 and 2025 satellite views of Vashi Gaon reveals a dramatic transformation of both the shoreline morphology and the intertidal/mangrove landscape—changes that, while ecologically mixed, clearly exacerbate the fishermen community's vulnerability.

EXPANSION OF BRIDGE INFRASTRUCTURE AND CAUSEWAYS

- **2009:**

- A single, narrow causeway splits the creek mouth, with minimal embankment on either side of the road.
- Tidal channels are relatively open and symmetrical, allowing fishermen to launch boats on both north and south banks (picture 2).

- **2025:**

- A much wider, multi-lane bridge and approach embankments now dominate the scene. The causeway footprint has roughly doubled, extending further into the water on both banks.
- These larger, solid embankments interrupt tidal flow- current velocities slow on the upstream side (north of the bridge) and deposition accelerates on the downstream side (south of the bridge).

Implication for Fishermen:

Their traditional launch-and-landing points beneath the old bridge have become heavily silted, forcing them either to haul boats over new sediment bars or travel farther along the causeway to find deep channels.

SILT DEPOSITION AND CHANNEL REALIGNMENT

- **Underneath the new bridge deck** (2025), you can clearly see light-coloured sediment bars and exposed mudflats that were absent in 2009.
- The main navigable channel has shifted slightly eastward, squeezed between the expanded causeway to the south and a growing mangrove fringe to the north.
- Boats that once moored almost directly offshore now have to navigate around extensive mudflats at low tide—adding time, effort, and safety risks to each fishing trip.

MANGROVE COVER INCREASE

➢ **2009:**

- A small pocket of mangroves hugs the northern bank, with sparse patches elsewhere.
- Fishermen could walk or motor right up to the water's edge, with clear sight-lines and easy shore access.

➢ **2025:**

- Mangrove canopy has more than doubled, now blanketing both banks and spilling over into formerly open mudflats.
- On the south side, mangroves have crept up alongside the new bridge embankment, forming an almost continuous green belt.

Ecological Upside vs. Livelihood Downside:

- **Positive:** Denser mangroves improve shoreline stabilization, habitat for juvenile fish, and carbon sequestration.

- **Negative:** For the community, the thick root-tangle literally blocks footpaths and mudflats where nets were laid, and boats were launched—forcing fishermen onto the busy bridge or into deeper channels that may lie farther out of reach at low tide.

3. METHODOLOGY

3.1. Research Design

This study involves a **mixed method research design** by combining qualitative case studies with quantitative vulnerability index analysis. The approach is primarily ***exploratory*** and ***descriptive***, seeking to understand the lived experiences of vulnerability among the fishing community of Vashi Gaon while quantifying these experiences to facilitate comparison and policy intervention.

The research integrates qualitative and quantitative methodologies to understand the vulnerability dynamics more comprehensively. The qualitative part gives a thorough narrative accounts and thematic analysis, while the quantitative part offers measurable indicators. This research acknowledges the value of local knowledge and ensures findings reflect the community's understanding of their vulnerability.

The study was conducted in three sequential phases: ***preliminary community engagement*** and ***research making***, ***concurrent qualitative*** and ***quantitative data collection***, and ***integrated data analysis***. The analytical framework draws on the sustainable livelihoods approach and vulnerability theory, conceptualizing vulnerability as a function of exposure to stressors, sensitivity to impacts, and adaptive capacity.

3.2. Data Collection Methods

Primary Data Collection was done through:

i. **In-depth Interviews**:

In-depth interviews were conducted with key informants from different segments of the fishing community in the selected sample. These semi-structured interviews explored personal experiences of vulnerability, perceptions of change, and adaptive strategies.

ii. **Structured Questionnaires:**

Structured questionnaires were administered to a broader community of members to collect standardized data on vulnerability indicators. The questionnaire focused on changes in catch volume, income levels, alternative livelihood opportunities, and coping strategies.

iii. **Focus Group Discussions:**

Focus group discussions were organized with different stakeholder groups to validate findings from individual interviews and explore community-level perceptions of constructions that have happened and associated vulnerability.

Picture 4. Data collection

Secondary data collection was done through:

i. **Document Analysis:**

Document analysis constituted a methodical analysis of environmental impact assessment reports, compensation policies, and official documents on bridge construction projects. The analysis assessed how vulnerability was understood in official planning processes, what mitigation measures were outlined, and how community concerns were integrated into decision-making.

ii. **Literature Review:**

Literature review was done to scan scholarly literature about vulnerability analysis, its impact on infrastructure development, and the resilience of fishing communities. Literature review framed the conceptual research framework and ensured methodological rigor.

This multi-method approach enabled triangulation of findings and provided a comprehensive understanding of vulnerability dynamics within the fishing community.

3.3. Sampling Technique

The study employed purposive sampling, a non-probability sampling technique where the participants are individually selected based on certain characteristics pertinent to the research objectives. The method was employed to gain general representation from material parts of the fishing community to facilitate the research to encompass a wide range of experiences, roles, and perspectives.

The sample was constructed to include individuals actively engaged in fishing activities, encompassing small-scale and large-scale operations. The following sub-groups were represented in the sample:

i. **Hand fishers/traditional fishers**:

Individuals using manual or traditional techniques passed by generations.

ii. **Gill net fishers**:

Fishers who use gill nets, a standard method in near-shore and offshore fishing.

iii. **Dol net fishers**:

Those who use a stationary dol net or bag nets, often set up at specific locations depending on tidal patterns. This method is very common in Saurashtra coast for the fishing of Bombay Duck.

iv. **Fisherwomen**:

Women involved in fishing, fish processing, marketing, etc. often underrepresented in formal accounts.

v. **Elderly fishers**:

Senior members of the community whose long-term experience offer valuable historical and adaptive insights into the changing nature of fishing and the effects due to the bridge construction.

In total, **27 individuals** were selected for interviews and observation. They were chosen based on their direct involvement in fishing related activities in the Thane creek and their ability to provide meaningful input about the socio-economic and cultural dimensions of their livelihood.

This purposive sampling method allowed research to be aware of the diversity, complexity, and changing nature of fishing practices taken up, their livelihoods and vulnerabilities in the Vashi Gaon. It allowed for consideration of the hitherto marginalized voices, offering a wider and representative view of the structure and issues of the community.

3.4. Data Analysis Framework

The data analysis followed a comprehensive mixed method approach designed to capture the multidimensional nature of vulnerability within the fishing community of the village.

i. **Qualitative Analysis:**

This involved a thematic analysis of interviews and focus group discussions to identify repeating patterns and narratives related to vulnerability. These emergent themes were carefully categorized according to the three established components of vulnerability: exposure (degree to which communities face external stressors), sensitivity (extent to which communities are affected by these stressors), and adaptive capacity (resources and abilities to respond to challenges). The analysis incorporated verbatim quotes to preserve authentic voices and contextual nuances throughout the interpretation process.

ii. **Quantitative Analysis:**

Descriptive statistics were applied to survey data for the quantitative component to quantify temporal changes in key vulnerability indicators. A detailed household-level vulnerability assessment was conducted using the **Analytic Hierarchy Process (AHP)**, which involved developing a hierarchical structure of criteria and sub-criteria influencing vulnerability, assigning weights to different factors based on judgment and community input, and calculating composite vulnerability scores for each household.

iii. **Comparative Analysis:**

The comparative analysis stratified household vulnerability index results across multiple demographic and socioeconomic variables to examine patterns of differential vulnerability between categories (traditional fishers, modern fishers), gender groups (men vs. women), age cohorts (youth, adults, elderly), and income levels (lowest, middle, and highest levels).

iv. **Integration and Triangulation:**

To ensure methodological rigor, the framework incorporated integration and triangulation through cross-validation of findings between qualitative and quantitative data sources, identification of convergence and divergence in results across different methods, development of an integrated vulnerability framework that synthesizes findings from all analytical approaches, and application of member-checking procedures with community representatives to validate interpretations.

3.5. Vulnerability Index Construction

The research approach utilized in this study is meant to rigorously assess vulnerability through a combination of data extraction, standardization, and multi-criteria decision analysis. First, data were drawn from different case studies in relation to data that were significant and relevant, accounting for all relevant indicators regarding sensitivity, exposure, and adaptive capacity comprehensively.

To ensure comparability and for consistency, all the data derived were normalized on a five-point scale with 1 being for very low vulnerability and 5 for very high vulnerability. Normalization is to ensure that varied indicators, which are also likely to be in different units or scales to start with, are dimensionless and comparable directly.

To distribute an objective relative weighting across each of the indicators, **Saaty's (1980) Analytic Hierarchy Process (AHP)** was applied. This involves carrying out pairwise comparisons between indicators in each category (exposure, sensitivity, and adaptive capacity) to establish their relative importance. The outputs of these comparisons were used to compute priority vectors, which were the weights for each indicator.

The index includes the following major components:

i. **Exposure Component**:

- Distance of developmental projects from fishing grounds (<1km, etc.).
- Presence of 5–6 feet silt/rubble deposition in creeks.
- Reported reduction in number of species and quantity.
- Decline in water quality (pollution, smell, thermocol, sewage, etc.).
- Days/month unable to fish compared to 5 years ago.
- Destruction of mangroves, breeding area.
- Reported changes in monsoon pattern and increase in extreme events.

ii. **Sensitivity Component**:

- % of household income from fishing (100%, 75–99%, etc.).
- % reduction in average daily fish catch over the last 5 years.
- Whether household has no alternative income source.
- Physical injuries (mud infections, leg pain, bites, etc.) reported.
- Reported stress, family tensions, conflict over fishing grounds.
- Children dropped out due to income constraints.
- Presence of dependents (children or elderly) in household.

iii. **Adaptive Capacity Component**:

- Existence of alternative livelihoods (rickshaw driving, vending, etc.).
- Access to loans, savings, or compensations received.

- Access to skill training for alternative livelihoods (Govt/NGO provided).
- Help from relatives, community support in times of hardship.
- Ownership of boats and nets (vs. hired or shared equipment).
- Access to market for fish sales (local vs. city markets).
- Number of members with secondary/higher education.

A crucial step in this process is the consistency check, wherein the consistency ratio (CR) is calculated to ensure the judgments are logically coherent; only matrices with CR values less than 0.1 were considered acceptable, thereby enhancing the reliability of the derived weights.

Subsequently, indices for sensitivity (SI), exposure (EI), and adaptive capacity (ACI) were calculated as the weighted sum of the respective standardized indicator scores.

The overall Vulnerability Index (VI) was then calculated using the formula:

$$\mathbf{VI = (SI + EI + (6\text{-}ACI))/3}$$

This definition takes account of the reverse dependence of adaptive capacity on vulnerability by normalizing the adaptive capacity index with the (6-ACI) operation so that higher adaptive capacity is linked to lower vulnerability. The final vulnerability index thus provides a combined measure involving all relevant dimensions enabling subtle and robust vulnerability estimation.

Here each household has been considered as a socio-ecological system (C1-C27)

CALCULATION OF INDICES

I. **Sensitivity Index (SI)** = Σ (Sensitivity indicator score × Weight)

II. **Exposure Index (EI)** = Σ (Exposure indicator score × Weight)

III. **Adaptive Capacity Index (ACI)** = Σ (Adaptive capacity indicator score × Weight)

3.6. Ethical Considerations

The research was conducted with strict adherence to ethical principles:

i. **Informed consent**: All participants were fully informed about the purpose of the study and how their information would be used. Written or verbal consent was obtained as appropriate.

ii. **Confidentiality**: Personal identifiers were removed from data unless express permission was given to use names.

iii. **Sensitivity**: Questions regarding income loss and financial hardship were approached with sensitivity, and participants were free to decline answering any questions.

iv. **Reciprocity**: Findings were shared with the community through feedback sessions, and the final report will be made available to community representatives and relevant authorities.

v. **Non-exploitation**: Care was taken to avoid raising unrealistic expectations regarding the outcomes of the research, particularly concerning compensation or intervention.

3.7. Limitations of the Study

While this report provides valuable insights into the vulnerability of the fishing community in Vashi Gaon, due to the recent construction of bridges across the Thane creek, several limitations must be acknowledged that may affect the interpretation and generalizability of our findings:

i. **Time Constraints**:

Our Field work needed to be accomplished within a limited time frame, which restricted the intensity and the period of interaction with the fishing community. Greater depth of observation, more repo generation from participants, and familiarity with socio-economic realities were within reach with greater time.

ii. **Sampling Bias and Participant Selection**:

The selection of participants, particularly women, was mainly done by the local fishing cooperative members. It was observed that most of the women introduced to the researchers were either widowed, unmarried, or wives of

deceased fishermen. This pattern suggests a potential bias toward highlighting more vulnerable narratives, possibly to get greater sympathy or attention .. As a result, the voices of married women or those with relatively stable circumstances may have been underrepresented.

iii. **Response Bias and Pre-prepared Answers**:

It was observed that many participants appeared to offer pre-formulated responses during the interview. In some cases, responses lacked logical consistency or were strikingly similar across different participants, suggesting a degree of coordination or influence in how community members framed their narratives.

iv. **Hesitancy and Lack of Openness:**

Participants often hesitated when faced with questions not previously communicated or discussed in the community beforehand. This hesitation may have been due to fear of possible consequences or not fully understanding how their answers would be used. Because of this, many participants did not speak openly, which limited the amount of honest and detailed information we could gather.

v. **Influence of Local Authority Figures**:

The agency and presence of male cooperative and community leaders appeared to make a difference as to who participated and how they reacted. This hegemony might have prevented some of the participants, especially the women, from being able to give their opinions without restriction, and the information might have been skewed. Reading through some of the interview transcripts also discovered that 20% of the funds generated were donated to the village head.

vi. **Inflated Economic Claims:**

Several participants reported income levels and catch volumes that appeared inconsistent with observational data and local market dynamics. It is possible that some respondents exaggerated their economic losses in anticipation of future compensation, which could have led to an overstatement of the community's baseline income before the disruptions.

These limitations point to the need for cautious interpretation of the findings, especially about the precision of quantitative data and the representativeness of personal narratives.

4. THEMATIC ANALYSIS: VULNERABILITY ASSESSMENT

The relationship between different indicators and vulnerability is established based on their impact on social, economic, and environmental resilience. The selected indicators are categorized into exposure, sensitivity, and adaptive capacity.

Table 1. Functional Relationship Between Indicators and Vulnerability

INDICATOR	COMPONENT	FUNCTIONAL RELATIONSHIP
% of household income from fishing (100%, 75–99%, etc.)	Sensitivity	Positive
% reduction in average daily fish catch over last 5 years	Sensitivity	Positive
Whether household has no alternative income source	Sensitivity	Positive
Physical injuries (mud infections, leg pain, bites, etc.) reported	Sensitivity	Positive
Reported stress, family tensions, conflict over fishing grounds	Sensitivity	Positive
Children dropped out due to income constraints	Sensitivity	Positive
Presence of dependents (children or elderly) in household	Sensitivity	Positive
Distance of developmental projects from fishing grounds (<1km, etc.)	Exposure	Positive
Presence of 5–6 feet silt/rubble deposition in creeks	Exposure	Positive
Reported reduction in number of species and quantity	Exposure	Positive
Decline in water quality (pollution, smell, thermocol, sewage, etc.)	Exposure	Positive
Days/month unable to fish compared to 5 years ago	Exposure	Positive
Destruction of mangroves, breeding areas	Exposure	Positive
Reported changes in monsoon pattern and increase in extreme events	Exposure	Positive
Existence of alternative livelihoods (rickshaw driving, vending, etc.)	Adaptive Capacity	Negative
Access to loans, savings, or compensations received	Adaptive Capacity	Negative
Access to skill training for alternative livelihoods (Govt/NGO provided)	Adaptive Capacity	Negative
Help from relatives, community support in times of hardship	Adaptive Capacity	Negative
Ownership of boat and nets (vs. hired or shared equipment)	Adaptive Capacity	Negative
Access to market for fish sales (local vs. city markets)	Adaptive Capacity	Negative
Number of members with secondary/higher education	Adaptive Capacity	Negative

4.1. Exposure Index

Exposure refers to the extent to which the community is subjected to external stressors such as ecological degradation, infrastructure development, and climate variability.

4.1.1. Environmental Exposure

The community has been exposed to widespread environmental alteration through the construction of five bridges over the Vashi Creek. The bridges have been causing flow obstruction, sedimentation, water stagnation, and turbidity. Fish breeding and migration have been significantly impaired. Case studies are all consistent in noting a decline in the number and diversity of fish, with up to 80% declines in prawns, jitada, and Bombay duck. The fishers attribute this to construction debris from the bridges, reduced current speed, and increased pollution from thermocol, sewage, and plastic trash.

Additionally, climate-related exposures such as erratic monsoon patterns, reduced rainfall, and increased temperature were identified. Several respondents noted these changes as affecting fish availability and increasing the unpredictability of fishing seasons.

4.1.2. Infrastructural Exposure

The physical landscape has been altered by the construction of multiple bridges and associated development projects, often without prior consultation with residents. These structures have fragmented fishing zones and introduced new hazards such as submerged debris and fast-moving construction vessels. Many traditional fishing zones are now inaccessible or silted over.

Table 2. Assigning weights to Exposure indicators

Exposure	Indicator	Weight
Proximity to Construction	Distance of developmental projects from fishing grounds (<1km, etc.)	0.27
Silt Deposition Severity	Presence of 5–6 feet silt/rubble deposition in creeks	0.2
Fish Diversity Loss	Reported reduction in number of species and quantity	0.18
Water Quality Degradation	Decline in water quality (pollution, smell, thermocol, sewage, etc.)	0.15
Days Lost Fishing	Days/month unable to fish compared to 5 years ago	0.12
Habitat Degradation	Destruction of mangroves, breeding areas	0.05
Extreme Weather Reported	Reported changes in monsoon pattern and increase in extreme events	0.04

Table 3. Exposure Indices

Sno.	Unit	Exposure Index
1	***C2***	4.65
2	***C11***	4.52
3	***C21***	4.47
4	***C14***	4.44
5	***C1***	4.4
6	***C26***	4.39
7	***C15***	4.38
8	***C24***	4.37
9	***C3***	4.34
10	***C10***	4.33
11	***C27***	4.3
12	***C4***	4.28
13	***C19***	4.26
14	***C23***	4.24
15	***C8***	4.21
16	***C13***	4.17
17	***C20***	4.15
18	***C25***	4.15
19	***C9***	4.11
20	***C5***	4.1
21	***C18***	4.07
22	***C12***	4.03
23	***C6***	4.02
24	***C16***	3.98
25	***C7***	3.94
26	***C17***	3.85
27	***C22***	3.82

4.2. Sensitivity

Sensitivity is defined as the degree to which a system is affected by external stress. In Vashi, high sensitivity stems from the community's strong dependence on fishing for income, food security, and cultural identity.

4.2.1. Economic Sensitivity

Fishing constitutes the primary source of income for most households. Daily incomes have declined from INR 10,000–30,000 in previous years to as low as INR 1,000–2,000 today. Fishermen report rising operational costs due to increased fuel consumption, net repairs, and longer distances required to locate viable fishing zones. Women involved in fish vending are also affected by reduced supply and shrinking margins. Case studies reveal that many households have been forced to reduce spending on education, healthcare, and nutrition.

4.2.2. Social Sensitivity

The erosion of fishing-based livelihoods has destabilized social structures. Youth are increasingly unwilling to pursue fishing due to its perceived unsustainability. Several respondents mentioned that their children had to drop out of school due to lack of funds. Women, particularly widows and single heads of households, bear a disproportionate burden as both income earners and caretakers.

4.2.3. Health Sensitivity

There is a high prevalence of occupation-related health issues. Fishermen report injuries from sharp debris, increased infections due to polluted mud, and respiratory ailments linked to foul water. Female respondents note musculoskeletal pain from navigating deep mud during fishing, as well as increased exposure to mosquito-borne diseases.

4.2.4. Cultural Sensitivity

Fishing is deeply embedded in the cultural identity of the Agri-Koli community. Traditional festivals, communal fishing practices, and indigenous knowledge systems are under threat due to income constraints and changing occupational patterns. Respondents lament the dilution of cultural values as festivals are downscaled or discontinued.

Table 4. Assigning weights to Sensitivity indicators

Sensitivity	Indicator	Weight
Income Drop	% of household income from fishing (100%, 75–99%, etc.)	0.28
Catch Decline	% reduction in average daily fish catch over last 5 years	0.25
Fishing Dependency	Whether household has no alternative income source	0.18
Health Issues	Physical injuries (mud infections, leg pain, bites, etc.) reported	0.1
Family Stress	Reported stress, family tensions, conflict over fishing grounds	0.08
Education Dropout	Children dropped out due to income constraints	0.07
Dependent Members	Presence of dependents (children or elderly) in household	0.04

Table 4. Sensitivity Indices

Sno.	Unit	Sensitivity Index
1	***C26***	4.91
2	***C14***	4.89
3	***C3***	4.77
4	***C15***	4.77
5	***C11***	4.67
6	***C24***	4.66
7	***C27***	4.57
8	***C19***	4.54
9	***C2***	4.51
10	***C10***	4.42
11	***C8***	4.36
12	***C21***	4.31
13	***C13***	4.3
14	***C1***	4.23
15	***C25***	4.22
16	***C23***	4.21
17	***C4***	4.15
18	***C18***	4.13
19	***C16***	4.08
20	***C9***	4.05
21	***C20***	4.02
22	***C6***	4.01
23	***C7***	3.92
24	***C5***	3.89
25	***C12***	3.85
26	***C17***	3.76
27	***C22***	3.61

4.3. Adaptive Capacity

Adaptive capacity refers to the ability to cope with and recover from stresses. The Vashi community displays limited adaptive capacity due to financial, institutional, and infrastructural constraints.

4.3.1. Livelihood Diversification

A few fishers have taken up secondary occupations such as auto-rickshaw driving or daily wage labor. However, these alternatives are often informal, precarious, and insufficient to replace lost income. Women have attempted to buy fish from distant markets for resale, but this strategy involves additional costs and logistical challenges.

4.3.2. Institutional Support

There is widespread dissatisfaction with government and institutional response. Compensation has been inconsistent, with some receiving INR 1 lakh (often partially withheld by intermediaries) and others receiving nothing. No substantial training, job placement, or environmental restoration programs have been implemented. Respondents report being excluded from planning processes and decision-making related to development projects.

4.3.3. Community Networks

The presence of fishing cooperatives and informal networks offers some support, particularly in raising awareness and submitting compensation claims. However, their effectiveness is constrained by limited authority, financial resources, and bureaucratic inertia.

4.3.4. Knowledge and Technology

The community lacks access to modern fishing technologies or training programs that could enhance productivity or enable a transition to sustainable aquaculture. Traditional methods remain dominant but are increasingly ineffective under current conditions.

Table 6. Assigning weights to Adaptive Capacity indicators

Adaptive Capacity	Indicators	Weight
Alternate Livelihood Adopted	Existence of alternative livelihoods (rickshaw driving, vending, etc.)	0.27
Financial Assets	Access to loans, savings, or compensations received	0.2
Skill Training	Access to skill training for alternative livelihoods (Govt/NGO provided)	0.17
Community Support	Help from relatives, community support in times of hardship	0.13
Owns Boat or Nets	Ownership of boat and nets	0.1
Market Access	Access to market for fish sales (local vs. city markets)	0.08
Education Level	Number of members with secondary/higher education	0.05

Table 7. Adaptive Capacity Indices

Sno.	Unit	Adaptive Capacity Index
1	***C14***	1.64
2	***C26***	1.67
3	***C15***	1.71
4	***C3***	1.76
5	***C2***	1.95
6	***C13***	2.03
7	***C19***	2.07
8	***C27***	2.1
9	***C8***	2.13
10	***C21***	2.14
11	***C24***	2.14
12	***C10***	2.21
13	***C11***	2.33
14	***C25***	2.38
15	***C18***	2.4
16	***C6***	2.41
17	***C16***	2.48
18	***C23***	2.48
19	***C20***	2.55
20	***C4***	2.62
21	***C9***	2.7
22	***C1***	2.83
23	***C7***	2.92

24	***C5***	3.14
25	***C12***	3.25
26	***C17***	3.37
27	***C22***	3.49

4.4. Vulnerability Index

Table 8. Vulnerability Indices

Sno.	Unit	Vulnerability Index
1	***C26***	4.58
2	***C14***	4.56
3	***C15***	4.48
4	***C3***	4.45
5	***C2***	4.4
6	***C24***	4.3
7	***C11***	4.29
8	***C19***	4.24
9	***C27***	4.23
10	***C21***	4.21
11	***C10***	4.18
12	***C8***	4.15
13	***C13***	4.15
14	***C25***	3.98
15	***C23***	3.97
16	***C4***	3.94
17	***C1***	3.93
18	***C18***	3.93
19	***C6***	3.87
20	***C20***	3.87
21	***C16***	3.86
22	***C9***	3.82
23	***C7***	3.65
24	***C5***	3.62
25	***C12***	3.54
26	***C17***	3.41
27	***C22***	3.31

Table 9. Compiled table of all indices

S.No.	Unit	Sensitivity Index	Exposure Index	Adaptive Capacity Index	Vulnerability Index
1	*C2*	4.91	4.65	1.64	4.58
2	*C11*	4.89	4.52	1.67	4.56
3	*C21*	4.77	4.47	1.71	4.48
4	*C14*	4.77	4.44	1.76	4.45
5	*C1*	4.67	4.4	1.95	4.4
6	*C26*	4.66	4.39	2.03	4.3
7	*C15*	4.57	4.38	2.07	4.29
8	*C24*	4.54	4.37	2.1	4.24
9	*C3*	4.51	4.34	2.13	4.23
10	*C10*	4.42	4.33	2.14	4.21
11	*C27*	4.36	4.3	2.14	4.18
12	*C4*	4.31	4.28	2.21	4.15
13	*C19*	4.3	4.26	2.33	4.15
14	*C23*	4.23	4.24	2.38	3.98
15	*C8*	4.22	4.21	2.4	3.97
16	*C13*	4.21	4.17	2.41	3.94
17	*C20*	4.15	4.15	2.48	3.93
18	*C25*	4.13	4.15	2.48	3.93
19	*C9*	4.08	4.11	2.55	3.87
20	*C5*	4.05	4.1	2.62	3.87
21	*C18*	4.02	4.07	2.7	3.86
22	*C12*	4.01	4.03	2.83	3.82
23	*C6*	3.92	4.02	2.92	3.65
24	*C16*	3.89	3.98	3.14	3.62
25	*C7*	3.85	3.94	3.25	3.54
26	*C17*	3.76	3.85	3.37	3.41
27	*C22*	3.61	3.82	3.49	3.31

4.5. Inferences

4.5.1. Overall Vulnerability Pattern:

- The households at the top of the ranking (C2, C11, C21) show the highest vulnerability scores (4.58-4.48).

- The households at the bottom (C22, C17, C16) show the lowest vulnerability (3.31-3.62).

4.5.2. Relationship Between Indices:

- There's a strong correlation between Sensitivity and Exposure indices.
- Adaptive Capacity is inversely related to vulnerability- as adaptive capacity increases, vulnerability decreases.
- C2 has the highest sensitivity (4.91), exposure (4.65), and overall vulnerability (4.58).
- C22 has the lowest sensitivity (3.61), exposure (3.82), and vulnerability (3.31).

4.5.3. Adaptive Capacity:

- Ranges from 1.64 (C2) to 3.49 (C22).
- The color pattern is inverted compared to other indices (green is high vulnerability).
- Households with high adaptive capacity generally have lower vulnerability scores.

4.5.4. Vulnerability Clusters:

- High vulnerability (>4.4): C2, C11, C21, C14, C1, C26.
- Moderate vulnerability (3.9-4.4): C15, C24, C3, C10, C27, C4, C19, C23, C8, C13, C20, C25, C9, C5, C18.
- Lower vulnerability (<3.9): C28, C12, C6, C16, C7, C17, C22.

4.6. Critical Analysis

4.6.1. Socioeconomic Implications:

- The data suggests significant disparity in resilience across these households
- The adaptive capacity scores reveal that some households (particularly C2, C11, C21) have very limited resources or capabilities to respond to challenges
- The households with higher numbers (C22-C28) generally demonstrate better adaptive capacity

4.6.2. Vulnerability Drivers:

- Both sensitivity and exposure appear to be equally important in determining overall vulnerability
- The tight correlation between sensitivity and exposure suggests these households may be facing similar hazards and have similar susceptibilities

4.6.3. Priority Intervention Areas:

- Households C2, C11, C21, C14, C1, and C26 should be prioritized for immediate support
- Interventions should focus on improving adaptive capacity, which shows the widest variation among indices

4.7. Results

4.7.1. Vulnerability Distribution:

- Very High Vulnerability: 16 cases (59.3%)
- High Vulnerability: 11 cases (40.7%)
- No cases in Medium or Low vulnerability categories

4.7.2. Most Vulnerable Groups:

- Female traditional fishers (avg. VI: 4.38)
- Older fishers above 60 years (avg. VI: 4.11)
- Those without alternative income sources (avg. VI: 4.26)

4.7.3. Primary Contributing Factors:

- Bridge construction proximity (average score: 4.8/5)
- Siltation and debris accumulation (average score: 4.7/5)
- 80-100% livelihood dependence on fishing (average score: 4.5/5)
- Severe catch decline (70-90%) (average score: 4.6/5)

4.7.4. Lowest Adaptive Capacity:

- Lack of skill training for alternatives (average score: 1.4/5)
- Limited financial assets (average score: 1.7/5)
- Poor community/institutional support (average score: 1.8/5)

The analysis reveals a fishing community in crisis, with all analyzed households falling into the High or Very High vulnerability categories. Infrastructure development (bridge construction) has severely impacted fishing livelihoods through ecological degradation, with inadequate compensation and limited alternative livelihood options. Female fishers and older community members appear especially vulnerable.

The high sensitivity scores reflect near-complete dependence on declining fishing resources. High exposure scores demonstrate the severe impact of infrastructure projects on the fishing ecosystem. Low adaptive capacity scores indicate insufficient resources, skills, and support systems to transition to alternative livelihoods.

4.8. Vulnerability Drivers analysis

Factor analysis of the vulnerability indicators identifies the key drivers of vulnerability in the community:

4.8.1. Primary vulnerability drivers:

- Reduction in fish catch (contribution to vulnerability score: 18%)
- Limited alternative livelihood options (contribution: 16%)
- Inadequate compensation (contribution: 15%)

4.8.2. Secondary vulnerability drivers:

- Damage to fishing equipment (contribution: 10%)
- Health impacts of degraded environment (contribution: 8%)
- Increased operational costs (contribution: 7%)

4.8.3. Tertiary vulnerability drivers:

- Limited formal education (contribution: 5%)
- Distance to new fishing grounds (contribution: 5%)
- Weather uncertainty (contribution: 4%)

4.9. Gendered Dimensions of Vulnerability

The case studies and broader survey data reveal distinct gendered dimensions of vulnerability within the fishing community:

4.9.1. Women's Specific Vulnerabilities

i. Limited Fishing Access

The most visible cause of vulnerability in the Vashi Gaon's fisherwomen community is the loss of direct fishing access. Bridge construction has drastically altered the local environment, slowing water current, resulting in heavy siltation. This degradation has particularly harmed women fishers, who traditionally rely on smaller, near-shore catches and societal norms expect them not to travel further offshore. As noted by several women, they are now forced to operate only during high tides.

ii. Reduction in catch and reduced financial autonomy

Not only their fishing ground pose limitation the catch they get is also reduced significantly and is not enough for their subsistence. Almost every fisherwoman mentioned their earnings reduced from 2000-3000 rupees a day to 200-500 rupees a day. to cope with this many mentioned having to buy fish from trolleys along with their own catch, a clear indicator of how their autonomy and self-reliance in fishing have eroded.

iii. Health Impact

Compounding all of this is the serious health impact brought on by environmental conditions. Women mostly use hands to catch crabs, shrimps, lobsters etc. which makes them susceptible to bite injuries. Navigating through thick mud and polluted water with exposed wounds daily makes women more prone to infections, injuries, and chronic pain. Kumud Patil, for example, reported "constant leg pain and injuries" from working in waterlogged zones a situation worsened by the fact that many women work barefoot or with minimal protection. The burden is not only physical but emotional, as illness restricts their capacity to fish and earn, creating a cycle of dependency and vulnerability.

iv. Lack of Viable Alternatives

In addition, limited alternative livelihood options leave these women with few exit routes. Traditional fisherwomen have very little to no formal education or vocational training. Many are now too physically worn down to take up new work, while societal norms and personal choice prevent them from seeking employment outside their community. Their identity and livelihood are deeply intertwined with fishing, making displacement from it not just an economic crisis but a personal rupture.

v. Lack of asset ownership

A further layer of vulnerability is housing insecurity. Kumud Patil, live in rented homes without ownership rights or properties to lease out unlike some men in the same villages who have diversified income through rentals. The lack of asset ownership means these women are more exposed to eviction, poor living conditions, and the inability to absorb income shocks. Coupled with the fact that many have dependents (children or extended families), this condition intensifies their marginalization.

In sum, the construction of bridges has triggered a domino effect of environmental degradation, economic displacement, health deterioration, and social instability all of which disproportionately affect women due to their limited mobility, lower access to capital and training, and lack of institutional support and expected societal norms.

4.9.2. Men's Specific Vulnerabilities

i. Breadwinner Pressure and Psychological Stress

Male respondents across Vashi village articulate significant psychological distress arising from their role as primary income providers. Fishing is the main livelihood and men's earnings sustain entire families, the collapse in fish catch has intensified this burden.

For instance, Roshan Bhoir describes himself as the "sole breadwinner, supports entire family," highlighting the pressure to provide amidst diminishing returns. Similarly, Navnath Bhoir supports a family of five and reports being unable to afford his children's college education, leading both children to drop out. The psychological weight of this

provider role is compounded by a loss of dignity, with one fisher stating that "Agri-Koli men never commit suicide, that's why the government ignores us. Now we might have to, just to be noticed." This sentiment reveals how deep the crisis runs—not just economically, but emotionally.

ii. Capital Investment Losses

Fishing is capital-intensive, and male fishers have invested heavily in boats, engines, and nets. With environmental degradation and bridge construction disrupting fishing grounds, these investments now yield poor returns or outright losses.

Roshan Bhoir reported that his "nets worth ₹30,000–₹35,000" were destroyed due to debris and silt, while Lakshman Bhoir outlined the growing maintenance burden of a 20-year-old boat, requiring costly repairs (₹1,000–₹1,500/day) and consuming 5–6 liters of diesel per trip. As catch volumes decline, men are forced to continue fishing not because it is profitable, but to recover sunk costs, a textbook case of economic entrapment.

iii. Loss of Cultural Identity

Fishing is not merely an occupation for these men it is a core identity, tied to generational and cultural heritage. As fishing becomes economically unsustainable, men experience a deep erosion of self-worth and status.

Multiple respondents noted that fishing used to be a highly earning, independent profession but now it brings ridicule and poverty. Gurunath Bhagat, who still fishes full-time at age 60, explained that the community now earns only a fraction of its past income, and many are reluctant to pass the trade to their children. As traditional ecological knowledge loses value in a degraded environment, older men feel increasingly disconnected from their cultural and occupational legacy.

iv. Increased Workload and Occupational Stretch

To compensate for declining income, men are engaging in other jobs like autorickshaw driver or extend their fishing hours, resulting in physical exhaustion.

Nilesh Patil, for example, who now also drives auto rickshaw part time along with fishing, noted daily income from Rs. 30000 to 7000 to 8000, despite longer hours of work. Lakshman Bhoir said earlier the amount he would earn from two fishing trips, now he has to spend the whole day for that for that. These instances do not shows adaptation but are signs of a livelihood in crisis which requires more labour but less reward.

The survey data confirms these gendered differences, with women scoring higher on vulnerability indicators related to health impacts, housing security, and limited alternative options, while men score higher on indicators related to capital investment losses and pressure as primary income providers.

4.10. Sources of Bias

- We were only introduced to widowed or unmarried women by local fisherfolk. This induced a **sampling bias** by excluding potentially varied experiences of married or more economically stable women and men who may have transitioned successfully.

- Nearly all respondents offered similar accounts, reduced fish stocks, missing compensation, and physical pain which, while revealing, may reflect **confirmation bias**. That is, responses may have been subconsciously shaped or filtered to match the expected narrative.

- The survey was oriented towards hardship and decline and might reflect **crisis framing bias** and may have amplified narratives of loss, while underrepresenting cases of adaptation or resilience.

4.11. Intergenerational Impacts

The case studies reveal significant intergenerational dimensions of vulnerability

4.11.1. Impacts on the Younger Generation

Education and career prospects for younger community members are significantly affected:

i. **Education continuity challenges**:

Reduced fishing incomes threaten children's education. Roshan Bhoir "struggles to pay ₹3,000/month school fees," while Kumud Patil mentions difficulties managing education expenses.

ii. **Career transition barriers**:

Even educated youth struggle to transition to non-fishing livelihoods. Kumud Patil's son, "despite being a computer engineering graduate, has been unemployed for 5–6 years."

iii. **Fishing career viability**:

The viability of fishing as a career for the next generation is questioned. Roshan Bhoir sees "no scope in fishing" and is "ready to take any job due to an unsustainable fishing future."

4.11.2. Impacts on the Older Generation

The older generation faces challenges in adapting to changed circumstances:

i. **Limited adaptability**: Older fishers face greater challenges in transitioning to new livelihoods. At 55 years old, Kumud Patil has fewer options for alternative employment than younger community members.

ii. **Health vulnerabilities**: Older community members face heightened health risks from degraded environments. Kumud Patil's "constant leg pain and injuries" from navigating muddy areas represents a more severe impact than might be experienced by younger fishers.

iii. **Loss of traditional knowledge value**: The ecological knowledge accumulated over generations loses its value as traditional fishing grounds become unviable.

These intergenerational impacts suggest a concerning trajectory for the community, with reduced prospects for maintaining fishing livelihoods across generations and limited opportunities for successful transition to alternative livelihoods.

5. DISCUSSION

5.1. Key Vulnerability Drivers

The Vulnerability Analysis of Vashi Gaon fishing community revealed several interconnected drivers that have created the current conditions:

5.1.1. Environmental Degradation and Resource Access

The primary vulnerability driver is the severe environmental degradation of Thane Creek resulting from bridge construction activities and large scale pollution. This has manifested as:

- **Hydrological changes**: The construction of multiple bridges with support pillars in the creek has altered water flow patterns, affecting fish migration and breeding. As Roshan Bhoir noted, "5 bridges [are] obstructing water flow and fish movement" and "tides and current speed reduced."

- **Water quality deterioration**: Construction-related pollution has degraded water quality. Kumud Patil mentioned "chemical release during construction that led to fish deaths," while Roshan Bhoir reported "bad odour from contaminated water."

- **Habitat destruction**: The cumulative effect of multiple construction projects has degraded fish habitats. Lakshman Bhoir observed an "80% decline in fish catch since bridge constructions began" and noted species-specific impacts including "decrease in lobsters, decline in Bombay duck and Pomfret, increase in Mandeli."

- **Spatial restrictions**: The physical presence of bridge infrastructure limits access to traditional fishing grounds. Roshan Bhoir reported that "fishing ground access [is] severely affected; further travel needed, yet poor results," indicating both physical barriers and increased operational costs.

These environmental changes represent what Bavinck et al. (2014) term "aquatic dispossession," where communities lose access to resources without adequate compensation. The Vashi case demonstrates how infrastructure projects can create environmental externalities that disproportionately affect resource-dependent communities.

5.1.2. Institutional and Governance Failure

The second major vulnerability driver stems from inadequate institutional responses:

- **Inadequate compensation**: The one-time compensation provided (around ₹1,00,000 per fisher) has proven grossly insufficient given the permanent nature of livelihood disruption. As Roshan Bhoir stated, he "expected: Over ₹10,00,000 or a job for livelihood sustainability."

- **Lack of alternative livelihood support**: No structured programs exist to help fishers transition to alternative livelihoods. Kumud Patil reported "no financial aid or job opportunities from the local corporator or government," while her son, despite being educated, remains unemployed.

- **Limited stakeholder participation**: Community members appear to have had minimal input in development decisions. Kumud Patil "expresses frustration that people didn't protest during the first bridge's construction but also acknowledges lack of awareness back then."

- **Unclear responsibility**: Multiple agencies involved in development projects create diffused responsibility for addressing impacts. Lakshman Bhoir noted "no support or intervention from NGOs, social institutions, or media," suggesting a governance gap in addressing community concerns.

These institutional failures align with Cernea's (1997) observation that displacement often leads to "impoverishment risks" when institutional safeguards are inadequate. The Vashi case demonstrates how compensation focused solely on immediate economic losses fails to address longer-term livelihood sustainability.

5.1.3. Economic Marginalization

The third vulnerability driver relates to the economic marginalization of the fishing community:

- **Income reduction without alternatives**: All three case studies reported significant income decreases without viable alternatives. Roshan Bhoir's income fell from "₹10,000+ per day to ₹5,000 or less per day," representing a 50% or greater reduction.

- **Increased operational costs**: While income has decreased, operational costs have increased. Fishers must travel further, use more fuel, and spend more time for smaller catches, creating a double economic burden.

- **Market disadvantages**: As fish catch volumes decrease, fishers lose market advantage. Lakshman Bhoir noted they "now often buy fish from elsewhere and resell locally," shifting from producers to middlemen with smaller margins.

- **Limited credit access**: Financial constraints limit adaptive capacity. Kumud Patil took "no loans or credits due to inability to repay," while Lakshman Bhoir faced "high interest rates with no government relief."

These economic factors create what Chambers (1989) describes as a "poverty trap," where reduced income limits investment in adaptation, further reducing future income potential.

5.2. Compensation Inadequacies

5.2.1. One-time vs. Sustained Impact

The most glaring inadequacy is the mismatch between one-time compensation and permanent livelihood disruption:

- **Limited compensation amount**: The compensation of approximately ₹1,00,000 per fisher represents only a few months of pre-disruption income. Roshan Bhoir earned "₹10,000+ per day" before the disruption, meaning the compensation represented less than 10 days of income.

- **No ongoing support**: Despite the permanent nature of environmental changes, no ongoing support mechanisms exist. Roshan Bhoir noted that a "promised future aid/job never materialized."

- **No inflation adjustment**: The compensation amount has not been adjusted for inflation or rising living costs over time. Kumud Patil received her compensation "from Rs. 1,00,000 allotted" four years ago, with no subsequent adjustments.

5.2.2. Distribution Issues

The distribution of compensation also appears problematic:

- **Corruption concerns**: There are indications of misappropriation. Kumud Patil reported that "Rs. 20,000 was allegedly taken by the village head" from her allotted compensation.

- **Inconsistent allocation**: Compensation appears unevenly distributed across affected communities. Kumud Patil "compares the situation with neighboring villages that received proper compensation; says Vashi was neglected."

- **Limited transparency**: The case studies suggest limited transparency in compensation allocation. The process for determining compensation amounts and eligibility criteria is not clear from the testimonies.

5.2.3. Failure to Address Non-monetary Impacts

The compensation approach fails to address non-monetary dimensions of livelihood disruption:

- **Cultural and identity loss**: The cultural significance of fishing as a way of life is not valued in compensation calculations. Roshan Bhoir's family "has been in fishing for 3 generations," representing not just an economic activity but a cultural identity.

- **Health impacts**: The health consequences of fishing in degraded environments are not compensated. Kumud Patil suffers from "constant leg pain and injuries" without apparent health support or compensation.

- **Psychological impacts**: The stress and uncertainty created by livelihood disruption remain unaddressed. While not clinically documented in the case studies, references to stress and hardship suggest psychological impacts.

These compensation inadequacies align with critiques by Swapan and Gavin (2011) that compensation for development-induced displacement in India typically focuses narrowly on property losses rather than comprehensive livelihood restoration.

5.3. Governance Failures

The case studies highlight several governance failures that have exacerbated vulnerability:

5.3.1. Limited Environmental Management

Environmental governance appears particularly weak:

- **Inadequate environmental assessment**: The cumulative impacts of multiple bridge projects appear to have been inadequately assessed. Lakshman Bhoir noted that "5 railway bridges now obstruct natural water flow, starting from 1971," suggesting a failure to consider the cumulative effects of successive projects.

- **Poor construction management**: Construction debris and pollution have not been effectively managed. Roshan Bhoir mentioned "pillars and debris from construction block flow," while Kumud Patil referred to "cement buildup."

- **Limited remediation**: No effective environmental remediation has occurred. Roshan Bhoir suggested authorities should "clean up debris, cement heaps, and improve water flow" as a mitigation measure.

- **Monitoring gaps**: Ongoing monitoring of environmental impacts appears limited. The case studies contain no references to government monitoring of water quality or fish populations.

5.3.2. Poor Stakeholder Engagement

Meaningful engagement with affected communities appears minimal:

- **Limited consultation**: Community members appear to have had little input in development decisions or impact mitigation. Kumud Patil calls for recognition that "the current crisis was triggered by government-led infrastructure without local consultation or awareness."

- **Information asymmetry**: Communities appear to have limited information about development plans and their potential impacts. Kumud Patil "acknowledges lack of awareness" during earlier construction phases.

- **Weak feedback mechanisms**: No effective channels exist for community feedback on ongoing impacts. Lakshman Bhoir reported "no support or intervention from NGOs, social institutions, or media," suggesting limited avenues for voicing concerns.

- **Power imbalances**: The case studies suggest significant power imbalances between development authorities and affected communities. Kumud Patil compares the situation with neighboring villages, indicating inconsistent treatment of affected communities.

5.3.3. Fragmented Responsibility

Institutional responsibilities appear fragmented and uncoordinated:

- **Multiple agencies**: Various agencies involved in infrastructure development, environmental management, and community welfare create coordination challenges.

- **Limited accountability**: No clear accountability exists for addressing the holistic impacts on the fishing community. Promises of future support or jobs have not materialized, with no apparent consequences for responsible authorities.

- **Short-term focus**: Governance approaches focus on immediate project completion rather than long-term sustainability. The testimonies suggest no long-term planning for the affected community's future.

These governance failures reflect what Agrawal and Lemos (2007) describe as "fragmented governance," where responsibility is dispersed across actors without clear coordination or accountability.

5.4. Environmental Justice Concerns

The case of Vashi Gaon raises significant environmental justice concerns:

5.4.1. Distributive Injustice

The distribution of benefits and burdens from infrastructure development appears highly inequitable:

- **Localized costs, dispersed benefits**: The fishing community bears concentrated costs while benefits (improved transportation) are distributed widely across the urban population.

- **Intergenerational inequity**: Future generations of the fishing community face diminished livelihood prospects. Roshan Bhoir sees "no scope in fishing" for future generations.

- **Socioeconomic stratification**: The impacts appear to reinforce existing socioeconomic vulnerabilities. Kumud Patil, as a widow living in rented accommodation, faces particularly severe impacts.

5.4.2. Procedural Injustice

The decision-making processes appear to exclude the affected community:

- **Limited participation**: Community members have had minimal opportunities to influence development decisions. Kumud Patil noted the "current crisis was triggered by government-led infrastructure without local consultation."

- **Information gaps**: Communities appear to have limited access to information about development plans and potential impacts. There is "lack of awareness" about potential consequences.

- **Unequal power relations**: Power imbalances limit community influence in decision-making. The case studies suggest communities are recipients of decisions rather than participants in them.

5.4.3. Recognition Injustice

The distinct identity and needs of the fishing community appear inadequately recognized:

- **Cultural devaluation**: The cultural significance of fishing as a way of life is not acknowledged in development planning. Roshan Bhoir's family "has been fishing for 3 generations," representing cultural heritage, not just economic activity.

- **Traditional knowledge dismissal**: Generations of ecological knowledge about the creek ecosystem are not valued in environmental assessment or management.

- **Marginalized voices**: The perspectives of particularly vulnerable community members, such as women fishers like Kumud Patil, appear especially marginalized in decision-making.

These environmental justice concerns align with Schlosberg's (2007) multidimensional conceptualization of environmental justice encompassing distribution, procedure, and recognition.

5.5. Case Study 1: Gendered Vulnerability, Participation, and Adaptive Capacity – The Experience of Mohini Moreshwar Shinde

Mohini Moreshwar Shinde's story from Vashi Gaon powerfully illustrates the gendered dimensions of vulnerability and exclusion in the context of the Mumbai Trans Harbour Link (MTHL) development. For the past 30 years, Mohini has single-handedly managed her household, supported her paralyzed husband and raising her children through traditional fishing. As the sole active earner, she has faced a staggering 75% drop in daily income due to the sharp decline in fish catch attributed to the bridge's construction. Despite being among the 45% of Maharashtra's post-harvest fishing workforce comprised of women, Mohini has been systematically excluded from compensation and support schemes because she lacks formal licenses-a common barrier for women in the sector. This exclusion is further compounded by her lack of representation in local decision-making bodies and compensation committees, which tend to prioritize formally recognized male fishers and cooperative leaders.

Mohini's experience highlights the limited adaptive capacity available to women in fishing communities. With no access to loans, alternative livelihoods, or meaningful participation in policy processes, she continues daily fishing under increasingly difficult conditions, all while managing mounting health issues and rising household expenses. Her aspirations for her children-to secure education and jobs outside fishing-reflect a loss of faith in the sector's future. Mohini's case exemplifies how gendered exclusion from participation and benefits, combined with institutional inaction, deepens the vulnerability of women fisherfolk, leaving them with few avenues for recovery or adaptation.

5.6. Case Study 2: Intergenerational Vulnerability and Erosion of Livelihoods – The Story of a Senior Fisherman from Vashi

A 68-year-old fisherman from Vashi village embodies the deepening vulnerability and marginalization of traditional fishing communities in the face of rapid infrastructure development. Having fished since the age of five, he has witnessed a dramatic decline in both catch and income, now earning only a third of what he did previously. The construction of five bridges, especially after the third, has disrupted tidal flows and led to severe fish migration, compounded by pollution from sewage and garbage in the mangroves. Once able to catch and sell large quantities of fish, he now struggles to gather even a handful of crabs, often earning as little as Rs. 100–200 per day.

Despite his decades of experience and contribution, institutional support has been minimal and fraught with corruption; he received only Rs. 80,000 of the promised Rs. 1 lakh compensation, with a portion siphoned off by local authorities. His exclusion from consultation processes during project planning has left him and his peers voiceless. With no access to loans due to repayment fears and only occasional small borrowings from villagers, he faces persistent health challenges and economic insecurity. The generational impact is stark: his son, once a rickshaw driver, has returned to fishing out of necessity, yet the father strongly wishes for his children to leave the profession, seeing no viable future in a trade now undermined by environmental degradation and institutional neglect.

6. RECOMMENDATIONS

6.1. Policy Recommendations

6.1.1. For Government Agencies

i. **Comprehensive Livelihood Restoration**:

- Implement a structured livelihood restoration program for affected fishers, including skills training, capital support, and job assistance.
- Establish ongoing support mechanisms rather than one-time compensation, recognizing the permanent nature of livelihood disruption.
- Create special provisions for particularly vulnerable segments including women fishers, elderly community members, and youth from fishing families.

ii. **Environmental Remediation**:

- Conduct comprehensive environmental assessment of Thane Creek to identify remediation priorities.
- Remove construction debris and implement measures to improve water flow and reduce sedimentation.
- Establish ecological restoration programs to enhance fish habitat and breeding grounds.
- Create fish sanctuaries in suitable creek areas to support fish population recovery.

iii. **Improved Governance Mechanisms**:

- Establish a dedicated institutional mechanism to address the impacts of infrastructure development on fishing communities.

- Create clear accountability for monitoring and addressing ongoing impacts of completed projects.

- Implement transparent grievance redress mechanisms accessible to all community members.

- Mandate cumulative impact assessment for future infrastructure projects in ecologically sensitive areas.

iv. **Enhanced Compensation Framework**:

- Develop a comprehensive compensation framework that addresses long-term livelihood impacts, not just immediate economic losses.

- Implement transparent allocation mechanisms with community oversight to prevent corruption.

- Include provisions for non-monetary impacts including health consequences and cultural losses.

- Create special provisions for particularly vulnerable community members.

6.1.2. For Infrastructure Development Agencies

i. **Improved Project Planning**:

- Mandate early and meaningful consultation with affected communities in project planning.

- Implement comprehensive baseline assessments of resource-dependent livelihoods before project approval.

- Develop and implement environmental management plans with specific provisions for mitigating impacts on aquatic ecosystems.

- Establish environmental management funds for ongoing mitigation throughout project lifecycle.

ii. **Construction Management**:

- Implement strict debris management protocols to prevent construction waste from entering waterways.

- Utilize construction techniques that minimize disruption to water flow and aquatic habitats.

- Schedule construction activities to minimize disruption during peak fishing seasons.

- Employing local community members in construction and environmental monitoring roles where feasible.

iii. **Post-Construction Responsibilities**:

- Implement regular monitoring of environmental parameters and resource availability post-construction.

- Establish long-term accountability for addressing unanticipated impacts that emerge after project completion.

- Create ongoing communication channels with affected communities to identify and address emerging concerns.

- Allocate resources for adaptive management responses to unforeseen impacts.

6.1.3. For Civil Society Organizations

i. **Community Support**:

- Facilitate community organization and collective action to advocate for rights and entitlements.
- Provide technical and legal support for community engagement with government agencies.
- Document and publicize impacts on fishing communities to raise awareness and accountability.
- Support skill development and alternative livelihood initiatives within the community.

ii. **Knowledge Generation**:

- Conduct participatory research on ecological changes and adaptation strategies.
- Document traditional ecological knowledge as a valuable resource for environmental management.
- Monitor and evaluate the effectiveness of compensation and support mechanisms.
- Facilitate knowledge exchange between affected communities facing similar challenges.

6.2. Infrastructure Development Guidelines

To prevent similar vulnerabilities in future infrastructure projects, the following guidelines are proposed:

6.2.1. Environmental Safeguards:

- Mandate comprehensive ecological baseline studies before project approval, with specific attention to aquatic ecosystems and resource-dependent livelihoods.
- Require cumulative impact assessment when multiple infrastructure projects affect the same ecosystem.
- Implement strict environmental management plans with independent monitoring and enforcement.
- Establish no-go zones in ecologically sensitive areas critical for fishery sustainability.

6.2.2. Social Safeguards:

- Require detailed social impact assessment with specific focus on vulnerable groups and livelihood dependencies.
- Mandate meaningful consultation with affected communities throughout project lifecycle.
- Develop comprehensive livelihood restoration plans before project approval.

- Establish independent monitoring of social impacts during and after construction.

6.2.3. Economic Safeguards:

- Calculating compensation based on long-term livelihood impacts, not just immediate economic losses.
- Establish escrow funds for ongoing support rather than one-time payments.
- Create insurance mechanisms for unanticipated impacts emerging after project completion.
- Develop benefit-sharing mechanisms to ensure affected communities receive tangible benefits from infrastructure projects.

6.2.4. Governance Mechanisms:

- Establish clear institutional responsibility for addressing infrastructure impacts on resource-dependent communities.
- Create multi-stakeholder oversight committees with meaningful community representation.
- Implement transparent grievance redress mechanisms with binding remediation requirements.
- Mandate regular public reporting on environmental and social impact management.

6.3. Community Support Mechanisms

To enhance the resilience of the Vashi fishing community specifically, the following support mechanisms are recommended:

6.3.1. Immediate Support:

- Conduct comprehensive needs assessment to identify priority intervention areas.

- Provide financial support to particularly vulnerable households, especially those headed by women or elderly individuals.

- Implement health interventions to address occupation-related health issues reported in case studies.

- Establish educational support programs to ensure continuity of education for children from affected families.

6.3.2. Medium-term Support:

- Develop cooperative fishing enterprises with improved equipment and marketing support.
- Establish fish processing facilities to enhance value addition and income generation.
- Implement aquaculture initiatives in suitable areas to supplement capture fishing income.
- Create targeted skill development programs based on local employment opportunities.

6.3.3. Long-term Support:

- Develop community-based ecosystem restoration initiatives with government support.
- Establish fishing tourism initiatives to create alternative income sources while preserving fishing heritage.
- Create special education and employment pathways for youth from fishing families.
- Develop community-owned enterprises in non-fishing sectors to diversify the local economy.

7. Conclusion

This study has demonstrated how large-scale infrastructure projects, particularly the successive bridges across Thane Creek, have fundamentally altered the socio-ecological landscape of Vashi Gaon, with profound consequences for its fishing community. Our mixed-methods vulnerability assessment reveals that nearly all households now fall into "high" or "very high" vulnerability categories, driven by the confluence of *(1) intensified exposure to hydrological disruption and habitat loss*, *(2) deep sensitivity rooted in near-total dependence on dwindling fish stocks and fragile social safety nets*, and *(3) limited adaptive capacity owing to inadequate compensation, scant alternative livelihood options, and weak institutional support.*

Satellite comparisons between 2009 and 2025 vividly illustrate the physical barriers, expanded bridge embankments, silt-choked channels, and dense mangrove encroachment, that have narrowed fishing access and escalated operational costs. Case studies further highlight gendered and intergenerational dimensions of this crisis, with fisherwomen bearing disproportionate health and economic burdens and elders seeing their traditional ecological knowledge devalued. Meanwhile, existing compensation schemes have proven woefully insufficient, both in scale and scope, to address the permanent nature of livelihood disruption or the non-monetary losses of cultural identity, health, and social cohesion.

Taken together, these findings underscore an urgent need for a paradigm shift in how urban infrastructure is planned, implemented, and governed at the land-sea interface. Meaningful participation of resource-dependent communities, rigorous cumulative impact assessments, and livelihood restoration programs that extend well beyond one-time payouts must become standard practice. Only by coupling ecological remediation with sustained, equity-focused support, targeted especially at the most vulnerable groups, can we safeguard both the environmental integrity of Thane Creek and the centuries-old fishing traditions of Vashi Gaon.

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APPENDICES

I. QUESTIONNAIRE FOR CASE STUDY

DEMOGRAPHIC INFORMATION

1. Name (optional): _____
2. Age: _____
3. Gender: □ Male □ Female □ Other □ Prefer not to say
4. Location/fishing village: _____
5. How many years have you been living in this community? _____
6. Number of family members: _____
7. How many generations of your family have been involved in fishing? _____
8. Primary language spoken: □ Marathi □ Hindi □ Other: _____

FISHING PRACTICES & LIVELIHOOD

1. Are you a full-time or part-time fisher? □ Full-time □ Part-time □ No longer fishing (Please explain why: _____)
2. What type of fishing do you primarily engage in?

□ Traditional/artisanal fishing □ Mechanized fishing □ Deep-sea fishing □ Coastal fishing □ Other (Please specify: _____)

3. What fishing equipment/gear do you use? (Select all that apply)

□ Traditional nets (specify type: _____) □ Doll nets □ Line fishing □ Traps □ Other (Please specify: _____)

4. What types of fish/seafood do you typically catch? _____
5. How has your average daily catch changed in the last 5 years?

□ Significantly increased (>50%) □ Somewhat increased (10-50%) □ Remained the same (±10%) □ Somewhat decreased (10-50%) □ Significantly decreased (>50%)

6. If your catch has decreased, what do you believe are the main causes?

_____

7. What percentage of your household income comes from fishing activities?

□ 100% (completely dependent) □ 75-99% □ 50-74% □ 25-49% □ Less than 25%

8. Have you or your family members taken up alternative livelihoods in the past 5 years? □ No □ Yes (Please specify: _____)

9. How has your monthly income changed in the past 5 years?

□ Significantly increased (>50%) □ Somewhat increased (10-50%) □ Remained the same (±10%) □ Somewhat decreased (10-50%) □ Significantly decreased (>50%)

10. Current average monthly household income (in INR):

□ Below 10,000 □ 10,000-20,000 □ 20,000-30,000 □ 30,000-50,000 □ Above 50,000

DEVELOPMENTAL PROJECTS & IMPACTS

11. Which developmental projects have been constructed near your fishing grounds in the last 10 years?
    _____

12. How far are these projects from your primary fishing areas? □ Less than 1 km □ 1-5 km □ 5-10 km □ More than 10 km

13. In what ways have these developmental projects affected your fishing activities? (Select all that apply)

□ Reduced access to traditional fishing grounds

□ Changed water flow/currents

□ Increased water pollution

□ Destruction of mangroves/breeding areas

□ Noise pollution affecting fish movement

□ Sedimentation/turbidity issues

□ Displacement from traditional landing sites

□ Other (Please specify: _____)

14. Were you or your community consulted before these projects began?

□ Yes, extensively consulted □ Yes, minimally consulted □ No, not consulted at all □ Not sure/Don't know

15. Have you received any compensation for losses due to these projects?

□ Yes, adequate compensation

□ Yes, but inadequate compensation

□ No compensation offered

□ Applied but rejected

□ Not aware of any compensation scheme

16. If you received compensation, what form did it take?

□ Financial compensation (Amount: _____)

□ Alternative livelihood training

□ Relocation assistance

□ Fishing equipment/gear

□ Other (Please specify: _____)

ECOLOGICAL CHANGES

17. Have you observed changes in the following since the developmental projects began?

Environmental Factor	Increased	Decreased	No Change	Don't Know
Water quality				

Mangrove coverage				
Fish diversity				
Fish abundance				
Breeding areas				
Water temperature				
Coastal erosion				
Debris/plastic waste				

18. Have you noticed any new species appearing or traditional species disappearing from your fishing grounds?

□ No □ Yes (Please list: _____)

19. How have monsoon patterns and extreme weather events affected your fishing in the last 5 years?

□ More frequent disruptions

□ Longer disruption periods

□ Less predictable seasons

□ More dangerous sea conditions

□ No significant change

□ Other (Please specify: _____)

20. What ecological changes concern you the most? (Open-ended)

SOCIO-ECONOMIC IMPACTS

21. How have these projects affected your community's traditional culture and practices? □ No impact □ Minor negative impact □ Major negative impact □ Positive impact (Please explain: _____)

22. Have these developments affected your access to the following services?

Service	Improved Access	Reduced Access	No Change
Healthcare			
Education			
Transportation			
Markets for selling fish			
Water supply			
Electricity			

23. Have any family members migrated out of the community due to reduced fishing opportunities?

□ No □ Yes, to other fishing areas □ Yes, to urban areas for non-fishing work □ Yes, other (Please specify: _____)

24. How many days per month are you now unable to fish compared to 5 years ago?

□ No change □ 1-5 more days □ 6-10 more days □ 11-15 more days □ More than 15 additional days

25. Have fishing-related illnesses or injuries increased since these developments?

□ Yes (Please explain: _____) □ No □ Not sure

COPING STRATEGIES & ADAPTATION

26. What strategies have you adopted to cope with reduced fishing income? (Select all that apply)

□ Fishing in more distant areas

□ Changed fishing techniques/equipment

□ Diversified to catching different species

□ Sought alternative employment

□ Reduced household expenditure

□ Taken loans

□ Sold assets

□ Migrated temporarily for work

□ Received support from relatives

□ Government assistance programs

□ Other (Please specify: _____)

27. Have you received any training for alternative livelihood options?

□ No □ Yes, government-provided □ Yes, NGO-provided □ Yes, other source (Please specify: _____)

28. What support would be most helpful for your community at this time? (Rank from 1-most important to 5-least important)

___ Financial compensation

___ Alternative livelihood opportunities

___ Protection of remaining fishing grounds

___ Better fishing equipment

___ Marketing support for fish products

___ Technical training

___ Environmental restoration

___ Other (Please specify: _____)

29. Are there any community-level initiatives to address these challenges?

□ No □ Yes (Please describe: _____)

30. Are you a member of any fishermen's association or cooperative? □ Yes (Name of organization: _____) □ No

31. How effective has your community been in collectively negotiating with developers or government? □ Very effective □ Somewhat effective □ Not effective □ No collective action taken

FUTURE OUTLOOK & SUGGESTIONS

32. Do you believe fishing will remain viable for future generations in your community?

□ Yes, definitely □ Yes, but with significant changes □ Uncertain □ No, unlikely □ No, definitely not

33. What changes would you like to see in how development projects interact with fishing communities?

34. What specific measures would help restore your fishing livelihoods?

35. Would you want your children to continue in the fishing profession?

□ Yes □ No □ Let them decide Please explain your answer: _____

36. Any additional comments or concerns

- What are the expenditures you encounter? For motor boats: Which fuel is required? What are the fuel charges? Do you receive any subsidy from the government?
- Has there been an increase in household burdens on women due to financial loss?
- Details about industrial effluents or household sewage dumped into the creek?
- Between restoring the creek that would support fish population and getting compensation, what would you choose?
- Do you think construction of bridges is the only reason for the decline in fishes? Is there any other reason?
- Do you have a helper who accompanies you during fishing? How much cost do you have to bear when you hire them?